3 Fig. 3.1 shows the variation with distance x of the gravitational potential ϕ between the surface of the Earth and the surface of the Moon. At point P, the gravitational potential is a maximum.

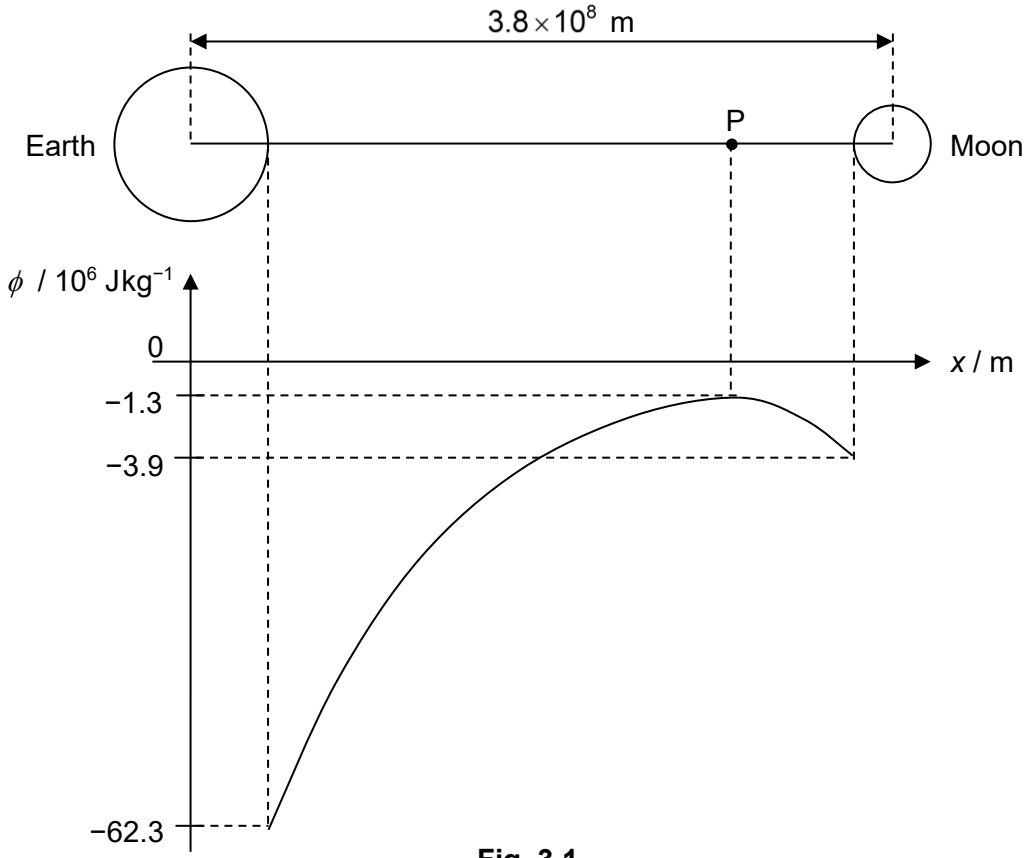

Fig. 3.1

The centres of both masses are separated by a distance of 3.8 x 10^8 m. The masses of the Earth and the Moon are 6.0 x 10^{24} kg and 7.4 x 10^{22} kg respectively.

(a) Define *gravitational potential* at a point and explain why the values of the potential shown in Fig. 3.1 are negative.

.....

.....

.....

.....

.....

.....

..... [3]

(b) **(i)** State how the resultant gravitational field strength at any point between the Earth and the Moon can be deduced from the graph in Fig. 3.1.

.....

..... [1]

(ii) Determine the distance from the centre of the Earth where the resultant gravitational field strength is zero.

distance = ... m [2]

(iii) On Fig. 3.2, sketch a graph to show the variation of the resultant gravitational field strength with distance *x* between the surface of the Earth and the surface of the Moon. Numerical values are not required.

Fig. 3.2

[2]

(c) A spacecraft is to be launched from the surface of the Earth to reach the surface of the Moon. Determine the minimum speed required to project the spacecraft from the surface of the Earth. Explain your answers clearly.

minimum speed = ... m s^{-1} [3]